SALSA-RL: Stability Analysis in the Latent Space of Actions for Reinforcement Learning

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Motivation

- Real-world control systems require **stable** behavior.
- Existing RL typically lack explicit mechanisms to ensure or analyze stability.

Highlights

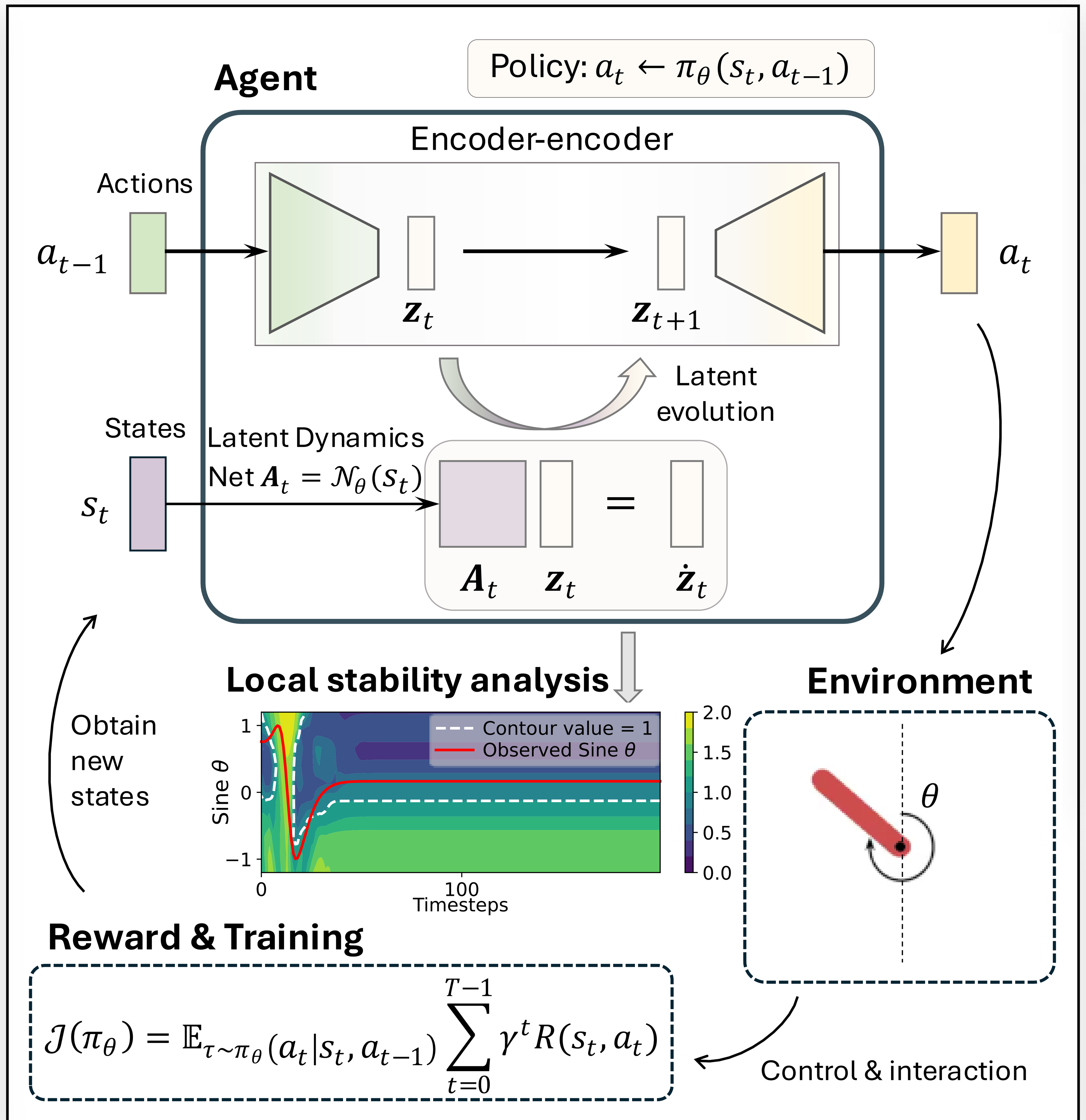
- SALSA-RL models actions as **latent** discrete dynamics.
- SALSL-RL analyzes **local stability** in latent action space, avoids high-risk regions and warns of control failures.
- Non-invasive integration with DRL training, no policy changes, same performance, improved interpretability.

Framework

- Encoder-Decoder (pretrained). $z_t = Encoder(a_t)$ $a_{t+1} = Dncoder(z_{t+1})$
- Latent Dynamics. $z_{t+1} = z_t + A_t z_t, \quad A_t = \mathcal{N}_\theta(s_t)$

Stability analysis for interpretability

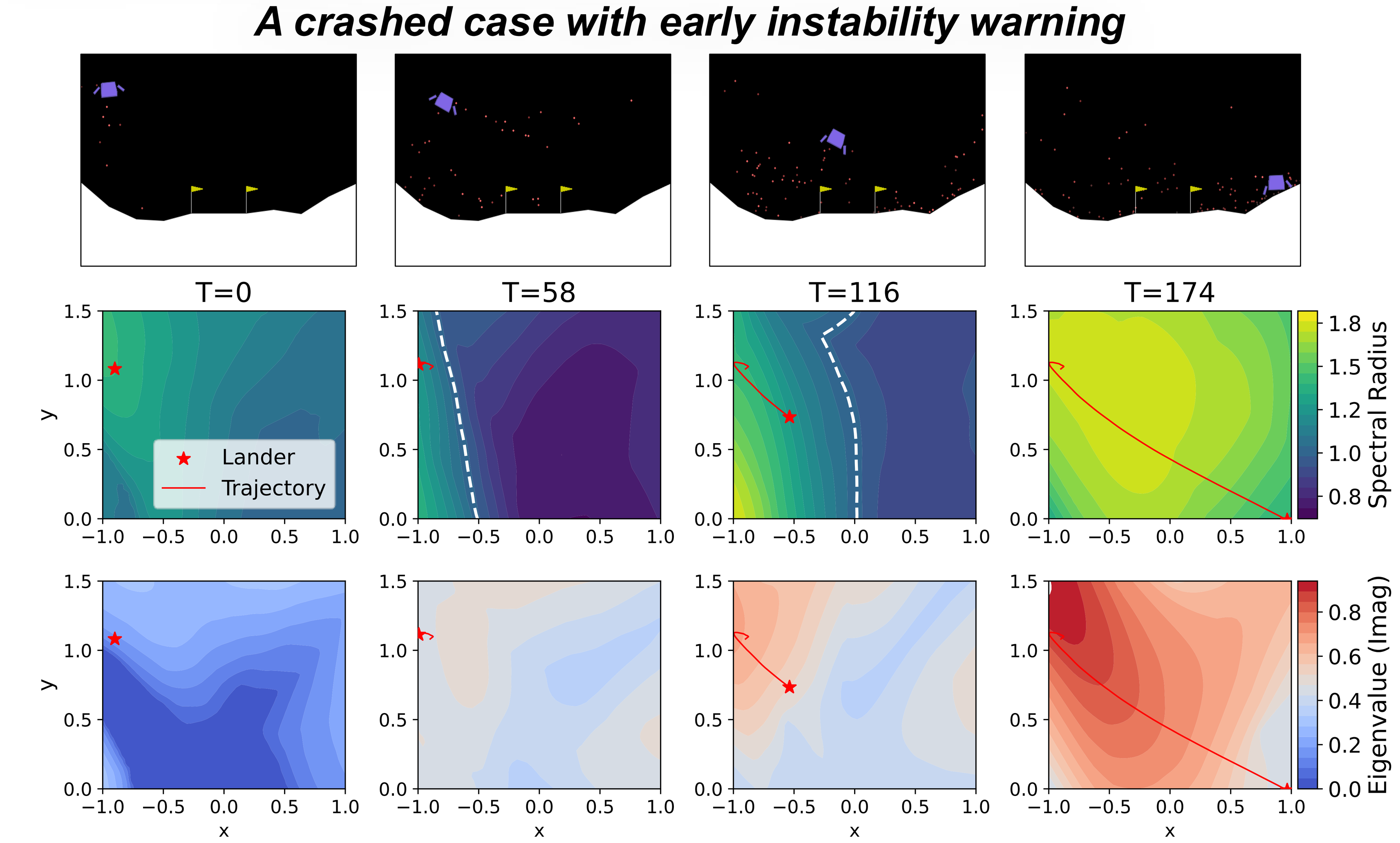
- **Eigenvalue**-based analysis of A_t. $\dot{z}_t = A_t z_t$
- Transient growth analysis (Kreiss constant η).
- Floquet analysis (periodic stability, Floquet exponents μ).



LunarLander control (floating objective)

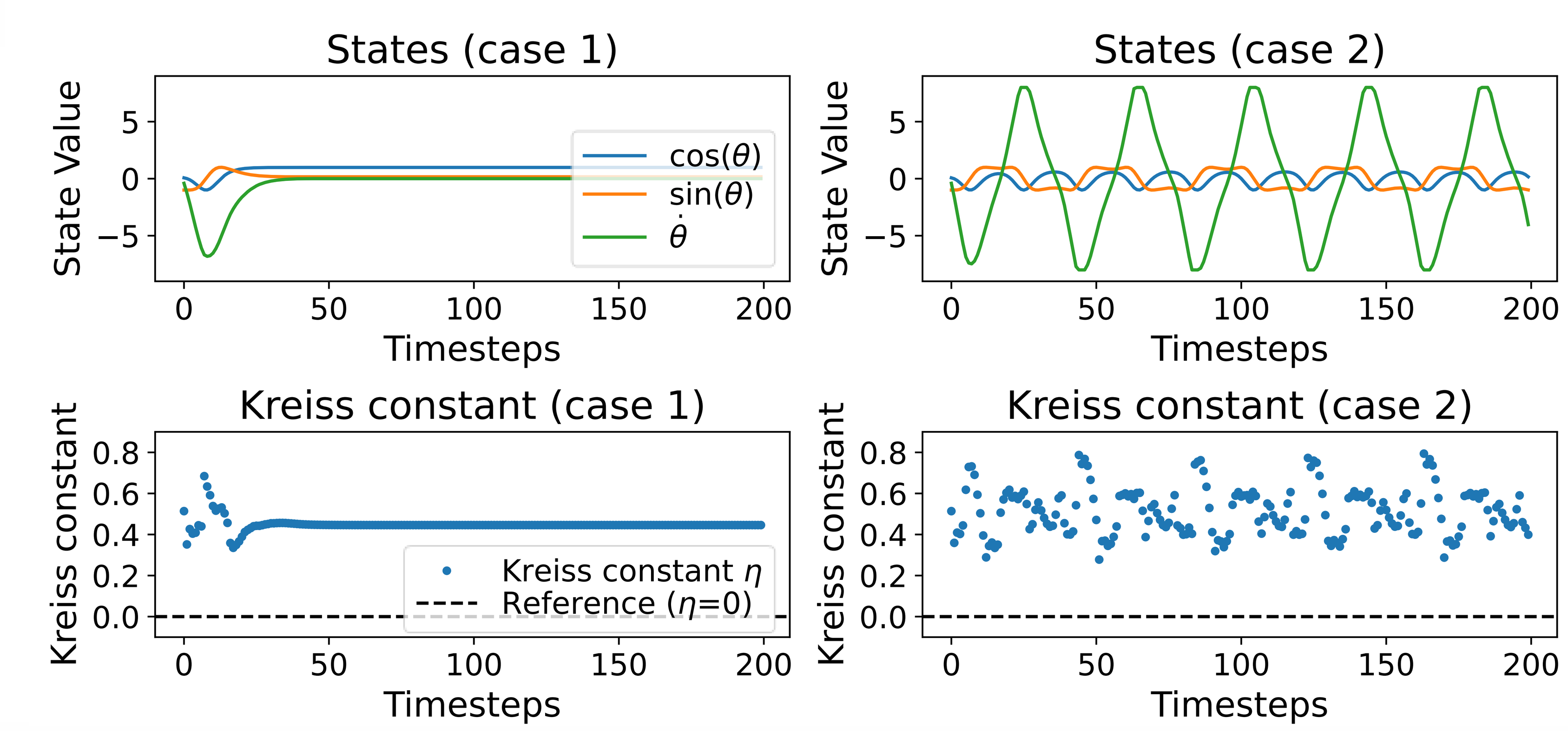
- Eigenvalue contours in 2D space. *Two actions: main/side engine thrusts.*
- **Early warnings of instability** before crash.

A crashed case with early instability warning



Transient growth & Floquet analysis

- Case 1 (success control) shows lower Kreiss constant η and minimal transient growth, unlike Case 2 (failed) with higher η and instability.
- Case 1 shows stable zero Floquet exponents, case 2 exhibits marginal growth and oscillation with $\mu_i = [0.96 + 0j, 0.02 + 0.04j, 0.02 - 0.04j]^T$.

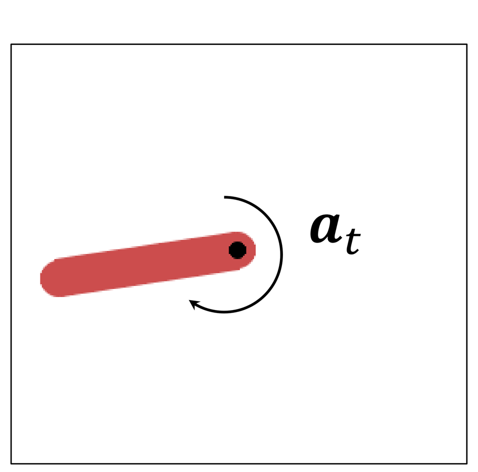


Conclusion

- SALSA-RL leverages latent action dynamics for local stability analysis.
- Local stability analysis reveals how control policies evolve safely by avoiding unstable regions.
- Transient growth & Floquet analysis uncover short-term instabilities and periodic failures missed by spectral analysis alone.

Pendulum control (continuous actions)

- Eigenvalue-based contours.
- Spectral radius & imaginary eigenvalue of time-dependent matrix $A_t = \mathcal{N}_\theta(s_t)$.
- Three regions:
  1) **Instability** $(T: 0-30)$,
  2) **Recovery** $(T: 30-150)$,
  3) **Control failure** $(T: 150-200)$.



Objective: swing pendulum upright